

Tejas Gokhale

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Assistant Professor; University of Maryland, Baltimore County

Current Position

University of Maryland, Baltimore County

- Assistant Professor, Department of Computer Science and Electrical Engineering 2023–
- Affiliate Faculty, Center for Artificial Intelligence 2024–
- Affiliate Faculty, Public Health Research Center 2026–
- Director, Cognitive Vision Group 2023–

Education

Arizona State University 2018–2023

Doctor of Philosophy (Ph.D.), Computer Engineering

Advisors: [Yezhou Yang](#), [Chitta Baral](#)

Thesis: Towards Reliable Semantic Vision

Carnegie Mellon University 2016–2017

Master of Science, Electrical and Computer Engineering

Advisor: [Aswin Sankaranarayanan](#)

Birla Institute of Technology and Science, Pilani 2011–2015

Bachelor of Engineering (Honours), Electronics and Instrumentation Engineering

Previous Affiliations

Johns Hopkins University Summer 2026

Visiting Scholar, [Human Language Technology Center of Excellence](#)

Microsoft Research Summer 2022

Research Intern, [Adaptive Systems and Interaction Group](#)

Mentors: [Hamid Palangi](#), [Besa Nushi](#), [Vibhav Vineet](#), [Eric Horvitz](#)

Lawrence Livermore National Laboratory Summer 2021, 2020

Research Scholar, [Machine Intelligence Group](#)

Mentors: [Rushil Anirudh](#), [Jay Thiagarajan](#), [Bhavya Kailkhura](#)

Arizona State University

Graduate Research Associate, [School of Computing and AI](#) 2018–2023

Graduate Teaching Associate, [School of Computing and AI](#) 2018–2020

Snap Research Summer 2018

Research Intern, [Computational Imaging Group](#)

Mentors: [Guru Krishnan](#), [Shree Nayar](#)

Carnegie Mellon University 2017–2018

Graduate Student Researcher, [Dept. of Electrical and Computer Engineering](#)

ST Microelectronics
Intern, [High Speed Links Group](#)

Fall 2014

Steel Authority of India Limited
Summer Intern, [Bhilai Steel Plant](#)

Summer 2013

Publications

My [Google Scholar](#) page has updates and citation information. Citations: 1200+, h-index: 19, i10-index: 23

Peer-Reviewed Conference Proceedings

- [C27] Shivanand Kundargi, Kowshik Thopalli, and Tejas Gokhale. “Sequentially Acquiring Concept Knowledge to Guide Continual Learning”. In: *Conference on Lifelong Learning Agents*. PMLR. 2026
[to-appear](#) CoLLAs 2026
- [C26] Sourajit Saha and Tejas Gokhale. “Zero-Shot Multimodal Retrieval with Multi-Scale Contextual Representations”. In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2026, pp. 20304–20324
<https://aclanthology.org/2026.acl-long.930/> ACL 2026
- [C25] Naresh Kumar Devulapally, Shruti Agarwal, Tejas Gokhale, and Vishnu Suresh Lokhande. “Latent Diffusion Unlearning: Protecting Against Unauthorized Personalization Through Trajectory Shifted Perturbations”. In: *Proceedings of the 33rd ACM International Conference on Multimedia*. 2025, pp. 11376–11384
<https://arxiv.org/abs/2510.03089> ACM Multimedia 2025
- [C24] Shaswati Saha, Sourajit Saha, Manas Gaur, and Tejas Gokhale. “Side Effects of Erasing Concepts from Diffusion Models”. In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2025, pp. 14991–15007
<https://arxiv.org/abs/2508.15124> EMNLP Findings 2025
- [C23] Nilay Yilmaz, Maitreya Patel, Yiran Lawrence Luo, Tejas Gokhale, Chitta Baral, Suren Jayasuriya, and Yezhou Yang. “VOILA: Evaluation of MLLMs For Perceptual Understanding and Analogical Reasoning”. In: *The Thirteenth International Conference on Learning Representations*. 2025
<https://arxiv.org/abs/2503.00043> ICLR 2025
- [C22] Sourajit Saha and Tejas Gokhale. “Improving shift invariance in convolutional neural networks with translation invariant polyphase sampling”. In: *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE. 2025, pp. 620–629
<https://arxiv.org/abs/2404.07410> WACV 2025
- [C21] Maitreya Patel, Naga Sai Abhiram Kusumba, Sheng Cheng, Changhoon Kim, Tejas Gokhale, Chitta Baral, et al. “Tripletclip: Improving compositional reasoning of clip via synthetic vision-language negatives”. In: *Advances in neural information processing systems*. Vol. 37. 2024, pp. 32731–32760
<https://arxiv.org/abs/2411.02545> NeurIPS 2024
- [C20] Agneet Chatterjee, Gabriela Ben Melech Stan, Estelle Aflalo, Sayak Paul, Dhruva Ghosh, Tejas Gokhale, Ludwig Schmidt, Hannaneh Hajishirzi, Vasudev Lal, Chitta Baral, et al. “Getting it right: Improving spatial consistency in text-to-image models”. In: *European Conference on Computer Vision*. Springer Nature Switzerland Cham. 2024, pp. 204–222
<https://arxiv.org/abs/2404.01197> ECCV 2024

- [C19] Agneet Chatterjee, Yiran Luo, Tejas Gokhale, Yezhou Yang, and Chitta Baral. “Revision: Rendering tools enable spatial fidelity in vision-language models”. In: *European Conference on Computer Vision*. Springer Nature Switzerland Cham. 2024, pp. 339–357
<https://arxiv.org/abs/2408.02231> ECCV 2024
- [C18] Agneet Chatterjee, Tejas Gokhale, Chitta Baral, and Yezhou Yang. “On the robustness of language guidance for low-level vision tasks: Findings from depth estimation”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 2794–2803
<https://arxiv.org/abs/2404.08540> CVPR 2024
- [C17] Maitreya Patel, Tejas Gokhale, Chitta Baral, and Yezhou Yang. “Conceptbed: Evaluating concept learning abilities of text-to-image diffusion models”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 13. 2024, pp. 14554–14562
<https://arxiv.org/abs/2306.04695> AAAI 2024
- [C16] Sheng Cheng, Tejas Gokhale, and Yezhou Yang. “Adversarial bayesian augmentation for single-source domain generalization”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, pp. 11400–11410
<https://arxiv.org/abs/2307.09520> ICCV 2023
- [C15] Man Luo, Zhiyuan Fang, Tejas Gokhale, Yezhou Yang, and Chitta Baral. “End-to-end knowledge retrieval with multi-modal queries”. In: *61st Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics (ACL). 2023, pp. 8573–8589
<https://arxiv.org/abs/2306.00424> ACL 2023
- [C14] Tejas Gokhale, Rushil Anirudh, Jayaraman J Thiagarajan, Bhavya Kailkhura, Chitta Baral, and Yezhou Yang. “Improving diversity with adversarially learned transformations for domain generalization”. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2023, pp. 434–443
<https://arxiv.org/abs/2206.07736> WACV 2023
- [C13] Maitreya Patel, Tejas Gokhale, Chitta Baral, and Yezhou Yang. “CRIPP-VQA: Counterfactual reasoning about implicit physical properties via video question answering”. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. 2022, pp. 9856–9870
<https://arxiv.org/abs/2211.03779> EMNLP 2022
- [C12] Tejas Gokhale, Abhishek Chaudhary, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. “Semantically Distributed Robust Optimization for Vision-and-Language Inference”. In: Association for Computational Linguistics, 2021, pp. 1493–1513
<https://arxiv.org/abs/2110.07165> ACL Findings 2022
- [C11] Tejas Gokhale, Swaroop Mishra, Man Luo, Bhavdeep Singh Sachdeva, and Chitta Baral. “Generalized but not Robust? Comparing the Effects of Data Modification Methods on Out-of-Domain Generalization and Adversarial Robustness”. In: *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, 2022, pp. 2705–2718
<https://arxiv.org/abs/2203.07653> ACL Findings 2022
- [C10] Neeraj Varshney, Pratyay Banerjee, Tejas Gokhale, and Chitta Baral. “Unsupervised Natural Language Inference Using PHL Triplet Generation”. In: Association for Computational Linguistics, 2021, pp. 2003–2016
<https://arxiv.org/abs/2110.08438> ACL Findings 2022

- [C9] Yiran Luo, Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. “To Find Waldo You Need Contextual Cues: Debiasing Who’s Waldo”. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics. 2022, pp. 355–361
<https://arxiv.org/abs/2203.16682> ACL 2022
- [C8] Man Luo, Arindam Mitra, Tejas Gokhale, and Chitta Baral. “Improving biomedical information retrieval with neural retrievers”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2022, pp. 11038–11046
<https://arxiv.org/abs/2201.07745> AAAI 2022
- [C7] Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. “Weakly supervised relative spatial reasoning for visual question answering”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, pp. 1908–1918
<https://arxiv.org/abs/2109.01934> ICCV 2021
- [C6] Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. “WeaQA: Weak Supervision via Captions for Visual Question Answering”. In: *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (2021), pp. 3420–3435
<https://arxiv.org/abs/2012.02356> ACL Findings 2021
- [C5] Pratyay Banerjee, Tejas Gokhale, and Chitta Baral. “Self-Supervised Test-Time Learning for Reading Comprehension”. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics. 2021, pp. 1200–1211
<https://arxiv.org/abs/2103.11263> NAACL 2021
- [C4] Tejas Gokhale, Rushil Anirudh, Bhavya Kailkhura, Jayaraman J Thiagarajan, Chitta Baral, and Yezhou Yang. “Attribute-Guided Adversarial Training for Robustness to Natural Perturbations”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 35. 9. 2021, pp. 7574–7582
<https://arxiv.org/abs/2012.01806> AAAI 2021
- [C3] Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. “Mutant: A training paradigm for out-of-distribution generalization in visual question answering”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. 2020, pp. 878–892
<https://arxiv.org/abs/2009.08566> EMNLP 2020
- [C2] Zhiyuan Fang, Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. “Video2commonsense: Generating commonsense descriptions to enrich video captioning”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. 2020, pp. 840–860
<https://arxiv.org/abs/2003.05162> EMNLP 2020
- [C1] Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. “VQA-LOL: Visual question answering under the lens of logic”. In: *European conference on computer vision*. Springer. 2020, pp. 379–396
<https://arxiv.org/abs/2002.08325> ECCV 2020

Peer-Reviewed Journals and Magazines

- [J1] Tejas Gokhale. “Towards robust visual understanding: A paradigm shift in computer vision from

recognition to reasoning”. In: *AI Magazine* 45.3 (2024), pp. 429–435
<https://doi.org/10.1002/aaai.12194>

AI Magazine

Peer-Reviewed Workshop Papers and Extended Abstracts

- [W7] Jessica RP Sutton, Thomas A Stanley, and Tejas Gokhale. “Using ground-based radar to investigate false detections in satellite precipitation”. In: *106th AMS Annual Meeting*. AMS. 2026
<https://ams.confex.com/ams/106ANNUAL/meetingapp.cgi/Paper/475997> AMS 2026
- [W6] Yiran Luo, Joshua Feinglass, Tejas Gokhale, Kuan-Cheng Lee, Chitta Baral, and Yezhou Yang. “Grounding Stylistic Domain Generalization with Quantitative Domain Shift Measures and Synthetic Scene Images”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 7303–7313
<https://arxiv.org/abs/2405.15961> CVPR 2024 Vision Datasets Understanding Workshop
- [W5] Tejas Gokhale. “Towards robust visual understanding: from recognition to reasoning”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 20. 2024, pp. 22665–22665
<https://ojs.aaai.org/index.php/AAAI/article/view/30281> AAAI New Faculty Highlights
- [W4] Tejas Gokhale, Joshua Feinglass, and Yezhou Yang. “Covariate shift detection via domain interpolation sensitivity”. In: *First Workshop on Interpolation Regularizers and Beyond at NeurIPS 2022*. 2022
<https://openreview.net/pdf?id=YkPjTHZDdm> NeurIPS 2022 Interpolation Workshop
- [W3] Kuldeep Kulkarni, Tejas Gokhale, Rajhans Singh, Pavan Turaga, and Aswin Sankaranarayanan. “Halluci-Net: Scene Completion by Exploiting Object Co-occurrence Relationships”. In: *CVPR Workshop on AI for Content Creation*. 2021
<https://arxiv.org/abs/2004.08614> CVPR 2021 AI for Content Creation Workshop
- [W2] Tejas Gokhale, Shailaja Sampat, Zhiyuan Fang, Yezhou Yang, and Chitta Baral. “Cooking With Blocks: A Recipe for Visual Reasoning on Image-Pairs”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2019, pp. 5–8
<https://arxiv.org/abs/1905.12042> CVPR 2019 Vision Meets Cognition Workshop
- [W1] Tejas Gokhale. “Vision beyond Pixels: Visual Reasoning via Blocksworld Abstractions”. In: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, 2019, pp. 6436–6437
<https://www.ijcai.org/Proceedings/2019/0907.pdf> IJCAI 2019 Doctoral Consortium

Technical Reports and Preprints

- [P4] Sarah Rastegar, Violeta Chatalbasheva, Sieger Falkena, Anuj Singh, Yanbo Wang, Tejas Gokhale, Hamid Palangi, and Hadi Jamali-Rad. “InfSplign: Inference-Time Spatial Alignment of Text-to-Image Diffusion Models”. In: *arXiv preprint arXiv:2512.17851* (2025)
<https://arxiv.org/abs/2512.17851> Preprint
- [P3] Ethan Wisdom, Tejas Gokhale, Chaowei Xiao, and Yezhou Yang. “Mole Recruitment: Poisoning of Image Classifiers via Selective Batch Sampling”. In: *arXiv preprint arXiv:2303.17080* (2023)
<https://arxiv.org/abs/2303.17080> Tech Report
- [P2] Tejas Gokhale, Hamid Palangi, Besmira Nushi, Vibhav Vineet, Eric Horvitz, Ece Kamar, Chitta Baral, and Yezhou Yang. “Benchmarking spatial relationships in text-to-image generation”. In: *arXiv preprint arXiv:2212.10015* (2022)
<https://arxiv.org/abs/2212.10015> Tech Report

- [P1] Maitreya Patel, Neeraj Varshney, Agneet Chatterjee, Tejas Gokhale, Yezhou Yang, Chitta Baral. Reliability-Checklist: Framework for Comprehensively Evaluating the Reliability of NLP Systems. <https://github.com/Maitreyapatel/reliability-checklist> Tech Report

Books and Monographs

- [B1] Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. *Advances in Multimodal Information Retrieval and Generation*. Synthesis Lectures on Computer Vision. Springer International Publishing, 2024. DOI: [10.1007/978-3-031-57816-8](https://doi.org/10.1007/978-3-031-57816-8)
<https://link.springer.com/book/9783031578151> ISBN: 978-3-031-57815-1

Chapters:

- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. “Transformer-Driven Models for Language, Vision, and Multimodality”. In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 11–34. ISBN: 978-3-031-57816-8. DOI: [10.1007/978-3-031-57816-8_2](https://doi.org/10.1007/978-3-031-57816-8_2)
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. “Multimodal Information Retrieval”. In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 35–91. ISBN: 978-3-031-57816-8. DOI: [10.1007/978-3-031-57816-8_3](https://doi.org/10.1007/978-3-031-57816-8_3)
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. “Multimodal Content Generation”. In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 93–134. ISBN: 978-3-031-57816-8. DOI: [10.1007/978-3-031-57816-8_4](https://doi.org/10.1007/978-3-031-57816-8_4)
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. “Retrieval Augmented Modeling”. In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 135–157. ISBN: 978-3-031-57816-8. DOI: [10.1007/978-3-031-57816-8_5](https://doi.org/10.1007/978-3-031-57816-8_5)

Ph.D. Dissertation

- [T1] Tejas Gokhale. 2023. Towards Reliable Semantic Vision. Order No. 30426752, Arizona State University. <https://www.proquest.com/docview/2813822780>

Intellectual Property

- [IP3] Tejas Gokhale, Sourajit Saha. *Zero-Shot Multimodal Retrieval with Multi-Scale Contextual Representations*. US Patent App. 64/045,450.
- [IP2] Hamid Palangi, Besmira NUSHI, Vibhav Vineet, Eric J Horvitz, Semiha E KAMAR EDEN, and Tejas GOKHALE. *Automated evaluation of spatial relationships in images*. US Patent 12,444,106. 2025 [\[url\]](#)
- [IP1] Tejas Gokhale, Rushil Anirudh, Jayaraman Thiagarajan, Bhavya Kailkhura, Chitta Baral, and Yezhou Yang. *Diversity using adversarially learned transformations for domain generalization*. US Patent App. 18/674,001. 2025 [\[url\]](#)

Funding

Sustainable Horizons Institute. <i>Sustainable Research Pathways (SRP) Program</i>	~\$30K 2026
DARPA SciFy (Scientific Feasibility) Program. <i>“Modular Reasoning using Hybrid Inferential Formalisms”</i>	(\$3.8M) 2024-27
UMBC Center & Institute Departmentally-Engaged Research (CIDER). <i>“Identification of Virga Precipitation Events”</i>	(\$50K) 2025-26
UMBC Cybersecurity Institute. <i>Cybersecurity Graduate Fellows Program</i>	(\$45K) 2025
UMBC Strategic Awards for Research Transitions (START) <i>“A Framework for Quantifying Typicality of AI-Generated Images”</i>	(\$25K) 2024-25
Maryland Procurement Office (via Johns Hopkins University) <i>“Modular Natural Language Understanding” (PI: Frank Ferraro)</i>	(~\$30K) 2024
UMBC Summer Research Faculty Fellowship (SURFF) <i>“Improving the Continual Learning Ability of Visual Recognition Systems via Targeted Unlearning”</i>	(\$8K) 2024
Microsoft Research Accelerate Foundation Models Academic Research <i>Cloud Computing and OpenAI Credits</i>	(\$20K) 2024
Google Cloud <i>Education Credits</i>	(~\$2.5K) 2023-24

Teaching

Instructor, UMBC

CMSC 472/672 Computer Vision [website]	Fall 2026, Fall 2025
CMSC 475/675 Neural Networks [website]	Spring 2026, Spring 2025
CMSC 491/691 Robust Machine Learning [website]	Fall 2024
CMSC 491/691 Computer Vision [website]	Spring 2024, Fall 2023
CMSC 898 Pre-Doctoral Candidacy Research	
CMSC 799 Master’s Thesis Research	
CMSC 699 Independent Study	
CMSC 499 Independent Study	

Graduate Teaching Associate, Arizona State University

CSE310: Data Structures & Algorithms	Spring 2020
CSE408: Multimedia Information Systems	Spring 2019
CSE110: Introduction to Programming,	Fall 2018

Guest Lecturer, Arizona State University

CSE598, Perception in Robotics	Spring 2022
CSE408, Multimedia Information Systems	Spring 2019

Student Instructor, BITS Pilani Goa Campus

CTE: Advanced Image Processing	Spring 2015
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Students

PhD

- Sourajit Saha (M.S. UMBC) Ph.D. CS [current], UMBC
 - * *Internships: SCALE 2026 & SCALE 2024 (at JHU HLTCOE)*
 - * *ACL Great Reviewer Award, 2026*
 - * *CVPR Outstanding Reviewer Award, 2026*
 - * *UMBC 3 Minute Thesis Finalist, 2026*
 - * *WACV 2025 Broadening Participation Scholarship*
 - * *GSA Professional Development Grant, 2025*
 - * *Lambda Research Grant Program 2025*
- Ziwei Zhang (M.S. USTC, China) Ph.D. CS [current], UMBC
 - * *CSEE Summer Research Fellowship, 2025*
- Shivanand Kundargi (B.S. KLE U., India) Ph.D. CS [current], UMBC
 - * *Internship: Lawrence Livermore National Laboratory, 2025*
 - * *UMBC Cyber Graduate Fellow, 2025*
 - * *CSEE Summer Research Fellowship, 2025 (declined)*
- Naren Sivakumar (M.S. UMBC) Ph.D. CS [current], UMBC
 - * *CSEE Summer Research Fellowship, 2026*
- Dylan Lang (M.S. ASU) Ph.D. CS [current], UMBC
- Sunny Bhati (M.S. Indiana U.) Ph.D. CS [discontinued], UMBC
 - * *CSEE Summer Research Fellowship, 2026*
- Jordan Turley (M.S. Harvard U.) Ph.D. CS [discontinued], UMBC

PhD (as Committee Member)

- Jacob Rubinstein (advisor: Don Engel) Ph.D. CS [current], UMBC
- Nilay Yilmaz (advisor: Yezhou Yang) Ph.D. CS [current], ASU
- Shubhasis Roy Dipta (advisor: Frank Ferraro) Ph.D. CS [current], UMBC
- Agneet Chatterjee (advisor: Yezhou Yang and Chitta Baral) Ph.D. CS [current], ASU
- Mark Jarzynski (advisor: Marc Olano) Ph.D. CS [current], UMBC
- Yiran Luo (advisor: Chitta Baral and Yezhou Yang) Ph.D. CS [current], ASU
- Maitreya Patel (advisor: Yezhou Yang and Chitta Baral) Ph.D. CS 2026, ASU
- Sheng Cheng (advisor: Yezhou Yang) Ph.D. CS 2025, ASU [\[dissertation\]](#)

MS Thesis (as Committee Member)

- Avi Donaty (advisor: Don Engel) M.S. CS 2026, UMBC
- Beamlak Bekele (advisor: Don Engel) M.S. CS 2026, UMBC
- Joey Mule (advisor: Riadul Islam) M.S. CE 2026, UMBC
- Naomi Angela Tack (advisor: Don Engel) M.S. CS 2024, UMBC

Other MS/PhD

- Research Assistant: Joey Mule Summer 2026

- Independent Study: Neel Patel Spring 2025, Fall 2024
- Independent Study: Shaswati Saha Ph.D. CS [current], UMBC
- Independent Study: Varun Magotra Spring 2024

Undergraduate

- Independent Study: Nicholas Harrell Spring 2025
- Independent Study: Alexander Shaner Spring 2025
- Visitors: Alexander Haibel 2025-2026
- Visitors: Tetevi Wilson, Dhanush Bharadwaj 2024-2025
- Visitors: Joey Mule, Luke Parrish 2023-2024
- UMBC CWIT Scholar: Chloe Wood 2024-25
- UMBC CWIT Scholar: Danielle Burton 2023-24

Teaching Assistants

- Aydin Ayanzadeh, CMSC 475/675 Neural Networks Spring 2026
- Yu Liu, CMSC 472/672 Computer Vision Fall 2025
- Ziwei Zhang, CMSC 475/675 Neural Networks Spring 2025
- Sourajit Saha, CMSC 491/691 Computer Vision Spring 2024
- Aidin Shiri, CMSC 491/691 Computer Vision Fall 2023

Ph.D. Mentees (at ASU)

- Maitreya Patel Ph.D. CS [current], ASU
- Agneet Chatterjee Ph.D. CS [current], ASU
- Nilay Yilmaz Ph.D. CS [current], ASU

MS (Thesis) Mentees (at ASU)

- Maitreya Patel M.S. CS 2022, ASU [\[thesis\]](#)
- Abhishek Chaudhary M.S. CS 2021, ASU [\[thesis\]](#)

Undergraduate Mentees (at ASU)

- ASU FURI Program: Mertay Dayanc B.S CS, 2020
- CS Capstone: Paul Butler, Jace Lord, Aashwin Ranjan, Sagarika Pannase, William Tith 2019-20

Academic Service

National Science Foundation

- Panel, IIS/III 2026, 2025
- Panel, GEO 2026
- Reviewer, GRFP 2025

Tutorial Chair, International Conference on Computer Vision (ICCV) 2025

Area Chair / Action Editor

- International Conference on Computer Vision (ICCV) 2025
- Advances in Neural Information Processing Systems (NeurIPS) 2024-25
- NeurIPS Position Papers Track 2025
- Winter Applications of Computer Vision (WACV) 2025-2026

- Association for Computational Linguistics (ACL) 2024
- North American Chapter of the ACL (NAACL) 2024
- Empirical Methods in Natural Language Processing (EMNLP) 2024

Session Chair

- Winter Applications of Computer Vision (WACV) 2025

Reviewer / Program Committee:

- Conference on Computer Vision and Pattern Recognition (CVPR) 2023–25
- International Conference on Computer Vision (ICCV) 2023
- European Conference on Computer Vision (ECCV) 2022–26
- Winter Conference on Applications of Computer Vision (WACV) 2021–24
- International Conference on Machine Learning (ICML) 2023–25
- Advances in Neural Information Processing Systems (NeurIPS) 2022–24
- International Conference on Learning Representations (ICLR) 2022–25
- AAAI Conference on Artificial Intelligence (AAAI) 2021–24
- AAAI Senior Member Presentation Track 2026
- Conference on Language Models (COLM) 2024
- Association for Computational Linguistics (ACL) 2021–26
- Empirical Methods in Natural Language Processing (EMNLP) 2021–23
- North American Chapter of the ACL (NAACL) 2021–23
- International Conference on Robotics and Automation (ICRA) 2019–2023
- International Conference on Intelligent Robots and Systems (IROS) 2022
- IEEE Robotics and Automation Letter (RA-L) 2020–24
- IEEE Transactions of Pattern Analysis and Machine Intelligence (T-PAMI) 2024–25
- ACM Multimedia 2025
- ACM Transactions of Computing for Healthcare 2024
- ACM Computing Surveys 2024
- Springer Machine Vision and Applications (MVAP) 2020
- Springer Book Proposals Reviewer 2024
- Mentor, AAAI Undergraduate Student Consortium 2024
- Award Committee, Best Student Abstract, AAAI 2024

Leadership:

- Organizer, Tutorial on Responsibly Building Generative Models [\[Website\]](#) ECCV'24
- Team Lead, Summer Camp for Applied Language Exploration (SCALE) 2024, JHU Human Language Technology Center of Excellence [\[Website\]](#) Summer 2024
- Organizer, Tutorial on Reliability of Generative Models in Vision [\[Website\]](#) WACV'24
- Organizer, Workshop on Open-Domain Reasoning under Multi-Modal Settings (ODRUM), [\[Website\]](#) [\[YouTube\]](#) CVPR'23
- Organizer, Workshop on Open-Domain Retrieval under Multi-Modal Settings (ODRUM), [\[Website\]](#) [\[YouTube\]](#) CVPR'22
- Organizer, Tutorial on Semantic Data Engineering under Multimodal Settings (SERUM) [\[Website\]](#) WACV'23
- Organizer, 2021 Frontiers of V&L Seminar Series, [\[Website\]](#), [\[YouTube\]](#) ASU

Misc:

- Lead Vocalist, CVPR House Band 2024, 2025

- Volunteer, 2019 Southwest Robotics Symposium, Tempe AZ SWRS 2019
- Volunteer, International Conference on Machine Learning 2020

University Service

University Service (at UMBC):

University-Level

- UMBC Strategic Planning Subcommittee on Research Computing and Infrastructure 2026–
- UMBC HPCF Governance, Subcommittee for CHIP-GPU 2025–
- Search Committee, Faculty Development Center Director 2026
- Reviewer, UMBC ORCA Internal Grants 2025
- Judge, UMBC Annual Research Symposium 2026
- PPR Seminar: Advances in Perception, Prediction, and Reasoning [\[Website\]](#)
- Faculty Learning Community, UMBC Faculty Development Center 2025–26
- Faculty Learning Community, UMBC Faculty Development Center 2024–25
- Member, Asian and Asian American Faculty Staff Council 2023–

College-Level

- Reviewer, COEIT Cybersecurity Research and Education Proposals Fall 2024
- Interviewer, COEIT Staff Searches 2025
- Reviewer, COEIT Student Summer Project Award Summer 2025
- Faculty Mentor, Center for Women in Technology 2023–
- Faculty Volunteer, COEIT Ph.D. Open House 2026, 2025, 2024
- COEIT Admitted Student Call Campaign 2026

Department-Level

- Course Development, CMSC 475/675: Neural Networks [\[Website\]](#)
- Course Development, CMSC 491/691: Robust Machine Learning [\[Website\]](#)
- Course Development, CMSC 472/672: Computer Vision [\[Website\]](#)
- Regular Graduate Course Proposal, CMSC 672: Computer Vision Approved
- Regular Undergraduate Course Proposal, CMSC 472: Computer Vision Approved
- CSEE Computer Science Undergraduate Committee 2025–
- CSEE Computer Science Graduate Admissions Committee 2023–25
- CSEE Publicity Committee 2023–25
- CSEE Computer Science Undergraduate Student Advisor 2023–
- Faculty Mentor, UMBC IEEE + Tau Beta Pi Open Lab Fall 2024
- CSEE Lightning Talks and Open House Fall 2025, Fall 2024, Fall 2023
- Reviewer, CSEE Research Day Spring 2024
- Interviewer, CSEE Faculty Candidates 2024, 2025

University Service (at ASU):

- Founder, Summer Vision Reading Group, ASU [\[Website\]](#)
- Course Development, CSE591: Frontier Topics in Vision & Language [\[YouTube\]](#) [\[website\]](#) Sp.2021, ASU
- Founding Advisor, ASU Machine Learning Club, ASU
- Award Reviewer, GPSA Teaching Award Reviewer ASU
- Mentor, Graduate Student Mentorship Program, ASU

- Project Mentor, CSE598 - Perception in Robotics, ASU Spring 2022
- Project Mentor, CSE576 - Natural Language Processing, ASU Fall 2018

Presentations

- (Invited Talk) CSEE Research Day, UMBC 04/2026
“Cognitive Vision: Intelligence as Concept Learning”
- (Invited Talk) Amazon Science 03/2026
“Concept-Based Understanding: A Tale of Four ‘R’s”
- (Invited Talk) Lawrence Livermore National Laboratory, CASC Seminar 03/2026
“Cognitive Vision: Intelligence as Concept Learning”
- (Keynote) Winter Applications of Computer Vision, Event Vision Workshop 03/2026
“Concept-Based Event Understanding: A Multi-Modal Perspective”
- (Invited Talk) International Conference on Computer Vision, AC Workshop 10/2025
“Active Data Pursuit for Robust Vision”
- (Invited Talk), UMBC COEIT Research Day 04/2025
“Cognitive Vision: Concepts, Contexts, and Semantics”
- (Invited Talk), UMBC Information Systems Seminar 11/2024
“Cognitive Vision: Concepts, Contexts, and Semantics”
- (Tutorial), European Conference on Computer Vision 10/2024
“Evaluation and Benchmarking for Text-to-Image Models”
- (Lightning Talk), IARPA Video-LINCS Proposers Day 02/2024
“Robust Visual Understanding: Knowledge-Guided and Multimodal Reasoning”
- (Tutorial), Winter Conference on Applications of Computer Vision 01/2024
“Challenges with Evaluation of Text-to-Image Models” [\[website\]](#)
- (Invited Talk), PRG Seminar, UMIACS (University of Maryland) 11/2023
“Robust Visual Understanding in the Multimodal Era”
- (Invited Talk) *“Towards Reliable Semantic Vision”* Spring 2023
- Rochester Institute of Technology (02/23)
 - Binghamton University (03/23)
 - University of Maryland Baltimore County (03/23)
 - Indiana University (03/23)
 - Case Western Reserve University (03/23)
 - Colorado School of Mines (03/23)
 - Temple University (04/2023)
- (Tutorial), Winter Conference on Applications of Computer Vision 01/2023
“Semantic Data Engineering for Robustness Under Multimodal Settings” [\[website\]](#)
- (Invited Talk) University of Illinois at Chicago 10/2022
“Robust Semantic Vision”

(Invited Talk) Microsoft Research <i>“Benchmarking Spatial Relationships in Text-to-Image Generation”</i>	10/2022
(Doctoral Consortium) CVPR, New Orleans <i>“Discovering Transformations for Generalization in Semantic Vision”</i>	06/2022
(Guest Lecture) Arizona State University CSE 598 <i>“Introduction to Generalization in Semantic Vision”</i>	03/2022
(Invited Talk) Arizona State University ML Club <i>“Robust Visual Understanding”</i>	09/2021
(Doctoral Consortium), IJCAI, Macao <i>“Vision Beyond Pixels”</i>	08/2019
(Tutorial) Telluride Neuromorphic Cognition Engineering Workshop, <i>“Reasoning about Objects and Actions via Block-Play”</i>	07/2019 [website]
(Invited) Birla Institute of Technology and Science (BITS Pilani) <i>“Deep Learning Methods in Imaging and Computer Vision”</i>	04/2018

Awards

- AAAFSC Inaugural Community Builder Award, UMBC	2026
- UMBC Career Connector Award	2026
- CVPR 2024 VDU Workshop, Best Paper Award	2024
- Research Excellence Award , ASU GPSA	2022
- Outstanding Mentor Award , ASU GPSA	2022
- NeurIPS Top Reviewer	NeurIPS 2022
- CVPR 2022 Doctoral Consortium	CVPR 2022
- ICLR Best Reviewer	ICLR 2022
- SCAI Doctoral Fellowship (ASU),	2022, 2021, 2020
- Engineering Graduate Fellowship, (ASU Engineering)	2023, 2020
- ASU GPSA Travel Award	for WACV 2023
- Graduate College Travel Award (declined)	WACV'23, CVPR'22
- Graduate College Travel Award (accepted)	ICCV'21, EMNLP'20, ECCV'20
- IJCAI 2019 Doctoral Consortium	IJCAI 2019
- Inducted, IEEE Eta Kappa Nu, Sigma Chapter	CMU, 2017
- National Talent Scholarship, National Council of Educational Research and Training (Govt. of India) 2007–2015	

Media

- DARPA’s AI is built to call BS on wild weapons claims Scientific American	04/2026
- UMBC team leads research into AI tools that can assess the feasibility of scientific claims UMBC News	04/2025
- Alum inspires next generation of computer vision researchers ASU Full Circle	10/2024

- [Frontiers of multimodal learning: A responsible AI approach](#)
Microsoft Research Blog 09/2023
- [CASC research in ML robustness debuts at AAAI conference](#)
News and Press, LLNL Computing 02/2021
- [HuggingFace and Intel release a solution for high-fidelity text and image consistency](#)
NetEase (163.com), China 04/2024

References

Available upon request